

XAI Visualization Tutorial: References Handout

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1 INTRODUCTION

This half-day tutorial was developed by the Making Visualization Scalable (MAVIS) for explaining machine learning classification models project (<https://gtr.ukri.org/projects?ref=EP%2FX029689%2F1>). The learning objectives were for attendees to understand: (1) How XAI requirements vary across stakeholders, (2) How visualization is used in XAI, (3) The types of issue and blooper that occur in XAI visualization, and (4) Scalability – how to make visualization effective for large/complex models. The tutorial was open to everyone, irrespective of their level of knowledge about XAI and visualization. This document contains the references used in each part.

2 PART 1: FUNDAMENTALS

The references for each topic were:

- What is XAI [27, 47]
- AI models [27]
- Stakeholders, XAI stages [46] and XAI tasks [1, 11, 27, 34, 45, 48]

3 PART 2: VISUALIZATION

The references for each topic were:

- Data types [8]
- Chart types used in XAI [11, 10]
- Tasks vs. visualizations:
 - Accuracy [38, 17, 41]
 - Feature importance [31, 25, 35, 24, 15]
 - Feature dependence [19]
 - Feature interaction [38, 31, 5, 6]
 - Example model predictions (unpublished work)
 - Misclassified [49, 23]

4 PART 3: SCALABILITY

The references for each topic were:

- Too many instances [29, 30, 18]
- Complex data (unpublished work)
- Data reduction [4]
- Aggregation [33, 22]
- Overplotting reduction [32, 14, 29, 50]

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- Dimensionality handling [9, 13] (one example is unpublished work)
- Pedestrian intention prediction (PIP):
 - From visual cues to human understanding in autonomous systems [12, 26, 36, 43, 21, 37, 20, 2, 39, 42, 28, 16, 7]
 - Methodology [3]
 - Visualization of model predictions [51, 44, 40, 3]
 - PIP using computer vision and machine learning [3]

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